Arista Access Node IND-ACC-65 Optical Transport Failure SOP

Node: IND-ACC-65 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_3791

Description

The occurrence of Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Arista Access Node IND-ACC-65 at the Indiranagar Exchange site is a critical failure domain that demands prompt attention. This failure domain arises from the inherent vulnerabilities of the Optical Transport Network (OTN), where any degradation or damage to the SFP/XFP transceivers or the underlying fiber infrastructure can lead to a cascade of errors and ultimately, node failure. The OTN operates at Layer 1 of the OSI model, leveraging protocols such as Optical Channel Transport Unit (ODU) and Optical Channel Payload Unit (OPU) to ensure reliable data transmission over optical fibers. However, the OTN's reliance on fragile SFP/XFP components and the susceptibility of these components to environmental stressors, such as temperature fluctuations, humidity, and contamination, make them prone to degradation and failure. Furthermore, the Indiranagar Exchange site, as a high-traffic location, is likely to experience increased stress on the network infrastructure, exacerbating the likelihood of OTN failures. Consequently, it is essential to promptly identify and rectify any issues arising from Optical Transport Failure to prevent network downtime and associated business disruptions.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Indiranagar Exchange site to facilitate physical access to the affected node.

- Verify the status of the redundant link to ensure it is functional and can assume traffic in the event of a failure.

- Perform a visual inspection of the SFP/XFP transceivers and surrounding infrastructure to identify any signs of physical damage or degradation.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss due to Environmental Stressors	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware sfp detail, Expected Output: SFP information, including temperature, voltage, and optical signal levels, Description: This command verifies the status of the SFP/XFP
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transceivers, including temperature, voltage, and optical signal
levels, which can help identify potential issues with the SFP/XFP
components.

Command: show interfaces transceiver detail, Expected Output:
Transceiver information, including optical signal levels and errors,
Description: This command provides detailed information about the
transceivers, including optical signal levels and errors, which can
help diagnose optical transport issues.

Command: show system environment, Expected Output: Environmental
information, including temperature and humidity levels, Description:
This command provides information about the environmental conditions
surrounding the node, which can help identify potential factors
contributing to Rx/Tx power loss.
```

Resolution Steps

Step 1: Step: Swap SFP/XFP Transceivers, Command: hw-module sfp slot 1 replace, Description: Swap the existing SFP/XFP transceiver with a new one to rule out any issues with the component.

Step 2: Step: Verify Fiber Integrity, Command: show interfaces transceiver detail, Description: Verify the integrity of the fiber connections and ensure they are secure and undamaged.

Step 3: Step: Perform Environmental Remediation, Command: show system environment, Description: Perform any necessary environmental remediation, such as ensuring proper temperature and humidity levels, to prevent Rx/Tx power loss.

Post-Check Verification

```
Command: show hardware sfp detail, Expected Output: SFP information,
including temperature, voltage, and optical signal levels,
Description: Verify that the SFP/XFP transceivers are functioning
correctly and that the optical signal levels are within acceptable
ranges.

Command: show interfaces transceiver detail, Expected Output:
Transceiver information, including optical signal levels and errors,
Description: Verify that the transceivers are functioning correctly
and that there are no errors or issues with the optical signal levels.

Command: show system environment, Expected Output: Environmental
information, including temperature and humidity levels, Description:
Verify that the environmental conditions are within acceptable ranges
to prevent Rx/Tx power loss.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Transceivers, Command: hw-module sfp slot 1 replace, Description: Restore the original SFP/XFP transceivers to their previous state in case the new ones fail to resolve the issue.

- Step: Revert Environmental Changes, Command: show system environment, Description: Revert any environmental changes made during the remediation process to their previous state.